CODE SPORT

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In this month's column, we discuss a medley of computer science interview questions.

In last month's column, we had discussed applying deep learning and NLP techniques to source code analysis. I had received queries from some of our readers for pointers on this topic. If you are interested in reading more on it, there is an excellent GitHub site, *https://ml4code.github.io/*, which contains the research papers, tools and software on this subject. This is an emerging area in deep learning and requires inter-disciplinary knowledge of both machine learning and software engineering. In this month's column, we discuss a few computer science interview questions, covering machine learning.

1. You are working on developing a part of a large image classification system of indoor household images taken from different cameras placed inside the house. Such a system can be part of a smart home project or for home security considerations. Since many of your customers have pet cats, your system needs to be able to recognise them. You have designed a machine learning system that can recognise cat images from non-cat images. Note that the household cameras being used in the setup are quite cheap, and can end up producing poor quality images due to variations in lighting, camera motion, etc.

   Since there are zillions of cat images on the Internet, you have decided to train your system using pet images from it, which are typically high quality. You have downloaded 100,000 labelled pet images from the Internet for training your cat recognition classifier. You have also obtained 5000 home-camera generated images, which can include cat images as well. You are planning to use 2500 of these 5000 images as a validation set for tuning the hyper-parameters. The remaining 2500 images will be used as a test set for final evaluation. After you have finished developing the initial model and are about to start doing a hyper-parameter tuning exercise as the next step, your friend says that your methodology is incorrect because your training set and validation set come from different data distributions and hence it is likely that your chosen model will perform badly on the test set. Do you agree with him? If no, explain why. If yes, explain how will you change your methodology? (Note that you cannot increase the amount of indoor home-camera image data, which is only 5000 images, of which a few hundred are cat images.)
2. You have been asked to set up a deep learning system, which can recognise faces of employees and send signals to open the entrance gate to the laboratory as they approach the gate. You have heard a lot about the advantages of end-to-end deep learning.

   In this case, if you decide to go with the end-to-end deep learning approach, you only provide the image of the employee approaching the door, and the neural network provides an open/close signal output which can be used to drive the entrance gate. The alternative is to employ a pipeline approach, wherein the first phase identifies the 'face part' from the image and puts a bounding box around it. The next phase does the face match recognition to check if he/she is a valid employee of the company. Which approach would you choose? Explain the rationale behind your choice.
3. You are given the problem of disambiguating entities in a given set of documents. For example, for the word 'Apple', you will need to disambiguate between the use of this word as a corporate entity vs as a fruit. A more complicated example is the following:

S1: The White House announced its intention to support the newly formed government in Vanaru.

S2: The New Year Party will be held at the White House, and will be hosted by Michelle Obama.

In sentence S1, White House is an entity that actually represents the American government.